

MODEL QUESTION PAPER

TED(10) 4108

(Revision 2010)

FIFTH SEMESTER DIPLOMA EXAMINATION IN TEXTILE TECHNOLOGY

FABRIC MANUFACTURE III

Time : 3 hrs

(maximum marks: 100)

PART - A

(Maximum marks = 10 marks)

Marks

- I
1. State the use of multiple box motions
 2. Mention the principle of terry motion
 3. Write the advantage of cam beat up
 4. List four important features of Air jet loom
 5. A shuttleless loom running with 720 MPM. If the picks per cm is 22, Calculate the production
- (5 x 2 =10)

PART – B

II. Answer any 5 questions – all questions carry equal marks

1. How multiple box motions are classified? Briefly explain
 2. What are the advantages of Automatic loom over ordinary shuttle loom?
 3. Compare conventional method of weaving and shuttleless weaving
 4. What precautions need to be taken in the preparation of warp and weft for shuttleless looms?
 5. Briefly explain the function of weft insertion elements in Air jet loom
 6. Mention the merits and de merits of water jet loom
 7. Find the time required to exhaust a weavers beam containing 1000 meters of yarn.
Picks per inch is 42 and the efficiency 75%. Speed 420 mpm
- (5×6 = 30)

PART – C

(Answer one full question from each module. Each carries 15 marks)

Module I

- III
- a) State the function of weft cutter (3)
 - b) Explain the working of non skip circular box motion with a neat sketch (12)

OR

- IV. a) What you mean by pick at will loom? (3)
- b) Explain the working of Dug dales Terry motion (12)

Module II

- V. a) Mention the principle of sulzer shuttleless weaving (3)
- b) Explain the working of rack and pinion type picking mechanism with neat sketch (12)

OR

- VI. a) State the merits of Rapier looms (3)
- b) Illustrate various stages of weft insertion in Sulzer shuttleless loom (12)

Module III

- VII. a) Mention the de merits of water jet looms (3)
- b) Explain the various stages of weft insertion in Maxbo airjet loom (12)

OR

- VIII. a) State the function of weft accumulator (3)
- b) Explain the working of water pump in a jet loom with a neat sketch (12)

Module IV

- IX. a) Find the length required to weave 100 meters of cloth. The take up and waste is 7% (3)
- b) A loom shed working with 120 automatic looms with an average speed of 680 MPM. An order placed urgently for 12000 meters gray cloth with 26 picks per cm. Average efficiency of this loom shed is 86%, maximum length of warp can wound on a beam is 800 meters, take up and waste is 7%. Find the time required to place the order. (12)

OR

- X a) Calculate the weight of the warp in a piece of cloth 40 meters long and 90 cm width with 0.5 cm selvedge on each side drawn 4 ends per dent. The number of ends ad picks per cm is 24 and 20 respectively. The count of warp and weft are 30 tex and 25 tex respectively. The regain of warp and weft is 10% and 8% respectively. (15)